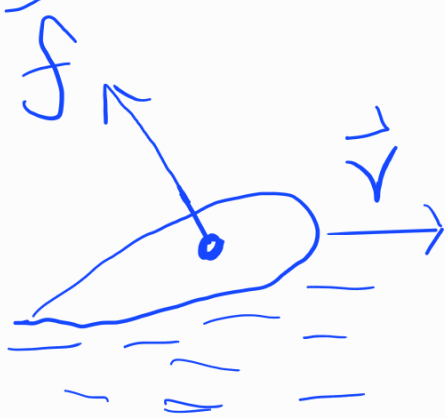


Air Resistance exists!

Drag: resistive force of material an object is moving through.

$$f \propto \vec{v} \text{ (velocity)}$$

Lift: sideways component of drag on an air plane wing



$$f_y \neq 0$$

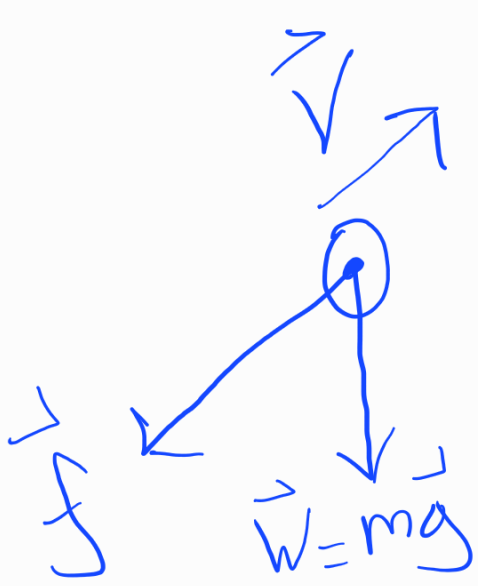
$$f_x \neq 0$$

Most cases: Drag & Velocity are in opposite directions



$$f_y = 0$$

$$f_x \neq 0$$



$$\vec{f} = f(\vec{v}) \hat{v}$$

$$\hat{v} = \frac{\vec{v}}{|\vec{v}|}$$

$$f(v) = \underbrace{bv}_{\text{linear}} + \underbrace{cv^2}_{\text{quadratic}} = f_{\text{lin}} + f_{\text{quad}}$$

f_{lin} : Comes from viscosity and linear size of object

f_{quad} : Due to density of medium and cross sectional Area of object

$$b \equiv \beta D, \quad c \equiv \gamma D^2$$

D : diameter of sphere

β & γ depend on medium properties

for a sphere Projectile @STP

$$\beta \approx 1.6 \times 10^{-4} \frac{\text{N.s}}{\text{m}^2}$$

$$\gamma \approx 0.25 \frac{\text{N.s}^2}{\text{m}^4}$$

Ratio of $\frac{f_{\text{quad}}}{f_{\text{lin}}}$ allows to

decide which can be neglected

$$\frac{f_{\text{quad}}}{f_{\text{lin}}} = \frac{cV^2}{bV^2} = \frac{\gamma}{\beta} DV$$

$$\approx \left(1.6 \times 10^3 \frac{\text{s}}{\text{m}^2}\right) DV$$

Ex 2.1

a) $D = 7 \text{ cm}$ & $V = 5 \text{ m/s}$

$$\frac{f_{\text{quad}}}{f_{\text{lin}}} \approx 600 \text{ [baseball]}, \text{ Need Quad}$$

$$f = -cV^2 \hat{V}$$

b) $D = 1 \text{ mm}$ & $V = 0.6 \text{ m/s}$

$$\frac{f_Q}{f_L} \approx 1 \text{ [raindrop]}, \text{ need both}$$

c) $D = 1.5 \mu\text{m}$, $V = 5 \times 10^{-5} \text{ m/s}$

$$\frac{f_Q}{f_L} \approx 10^{-7} \text{ [milli-kandrop]}, \text{ only need Linear}$$

$$f = -bV\hat{r} = -b\vec{V}$$

Reynolds Number:

$$R = DV \frac{\rho}{\eta}, \quad \rho = \text{density}$$

$\eta = \text{Viscosity}$

for Large R

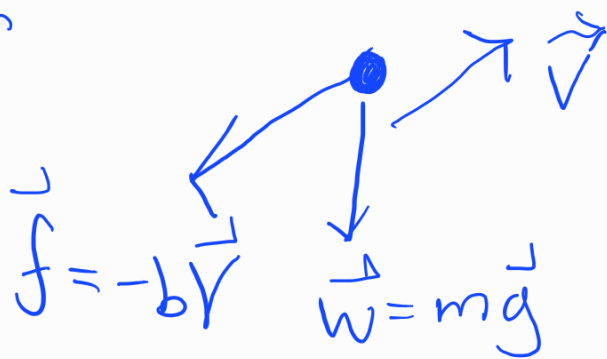
f_{quad} is dominate

for small R

f_{lin} is dominate

Linear Air Resistance

$$\vec{F} = m\ddot{\vec{r}}$$



$$m\ddot{\vec{r}} = m\vec{g} - b\vec{v}$$

$$m\dot{\vec{v}} = m\vec{g} - b\vec{v} \quad [\text{First Order differential Eqn.}]$$

$$\begin{aligned} m\dot{v}_x &= -bv_x \\ m\dot{v}_y &= mg - bv_y \end{aligned}$$

if $\vec{f} = \vec{f}_{\text{quad}}$

$$\begin{aligned} &= -c v^2 \hat{v} \\ &= -c v^2 \frac{\vec{v}}{|\vec{v}|} = -c v \vec{v} \end{aligned}$$

$$\begin{aligned} m\dot{v}_x &= -c \sqrt{v_x^2 + v_y^2} v_x \\ m\dot{v}_y &= mg - c \sqrt{v_x^2 + v_y^2} v_y \end{aligned} \quad \left. \vphantom{\begin{aligned} m\dot{v}_x &= -c \sqrt{v_x^2 + v_y^2} v_x \\ m\dot{v}_y &= mg - c \sqrt{v_x^2 + v_y^2} v_y \end{aligned}} \right\} \begin{array}{l} \text{Coupled} \\ \text{and} \\ \text{Complex} \end{array}$$

For Linear drag

$$\left. \begin{aligned} m\dot{V}_x &= -bV_x \\ m\dot{V}_y &= mg - bV_y \end{aligned} \right\} \begin{array}{l} \text{uncoupled} \\ \text{and} \\ \text{solved individually} \end{array}$$

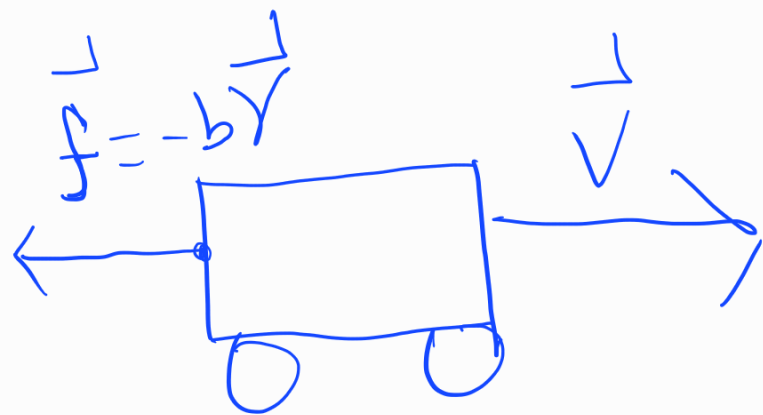
Horizontal motion w/ linear drag:

$$m\dot{V}_x = -bV_x, \text{ slows down, @ } t=0, x=0, V_x=V_{0x}$$

$$\dot{V}_x = -\frac{b}{m}V_x = -K V_x, K = \frac{b}{m},$$

$$V_x(t) = A e^{-Kt} = V_{x0} e^{-Kt} = V_{x0} e^{-t/\tau}$$

$$\tau = 1/K = m/b \text{ [for linear drag]}$$



$$V_x(t) = \dot{x} = V_{x0} e^{-t/\tau}$$

$$X(t) = \int_0^t \dot{X}(t') dt = X(t) - X(0)$$

$$X(t) = X(0) + \int_0^t V_{x0} e^{-t'/\tau} dt'$$

$$= 0 + [-V_{x0} \tau e^{-t'/\tau}]_0^t$$

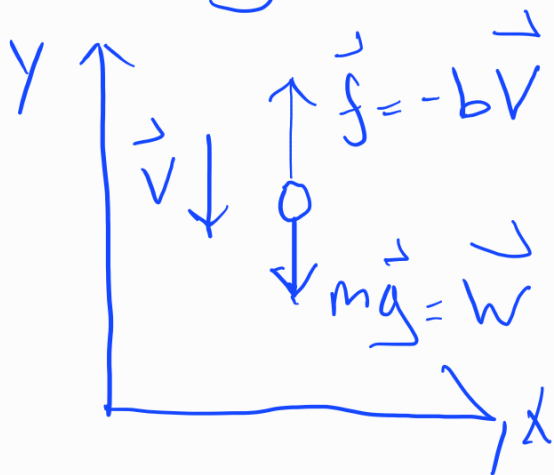
$$= [V_{x0} \tau - V_{x0} \tau e^{-t/\tau}], X_{\infty} = V_{x0} \tau$$

$$= X_{\infty} (1 - e^{-t/\tau})$$

$$X(t) \rightarrow X_{\infty} \text{ as}$$

$$t \rightarrow \infty$$

Vertical Motion with Linear Drag:



$$m\dot{v}_x = -bv_x = 0$$

$$m\dot{v}_y = mg - bv_y$$

$$m\dot{v}_y = mg - bv_y, \text{ for small enough } v_y \text{ eventually}$$

$$m\dot{v}_y = 0, \quad mg = bv_y,$$

$$V_{\text{ter}} = v_y = \frac{mg}{b}, \quad [\text{terminal speed}]$$

Ex 2.2

Millikan oil drop

$$D = 1.5 \mu\text{m} \quad \& \quad \rho = 840 \text{ kg/m}^3$$

$$b = \beta D, \quad \beta = 1.6 \times 10^{-4}$$

$$V_{\text{ter}} = \frac{mg}{b}, \quad m = \rho V = \rho \frac{\pi}{6} D^3 = \frac{4\pi}{3} r^3$$

$$V_{\text{ter}} = \frac{\rho \pi D^3 g}{\beta D} = \frac{\rho \pi D^2 g}{\beta}$$

$$= 6.1 \times 10^{-5} \text{ m/s}$$

for $D = 0.2 \text{ mm}$

$$V_{\text{ter}} = 1.3 \text{ m/s}$$

We can rewrite:

$$m\dot{v}_y = mg - bv_y \quad V_{ter} = \frac{mg}{b}$$

$$\text{as } m\dot{v}_y = -b(v_y - V_{ter})$$

$$\text{set } u = (v_y - V_{ter})$$

$$m\dot{u} = -bu \quad (V_{ter} \text{ is constant})$$

$$u = Ae^{-t/\tau}, \quad \tau = 1/\kappa$$

$$v_y - V_{ter} = Ae^{-t/\tau}$$

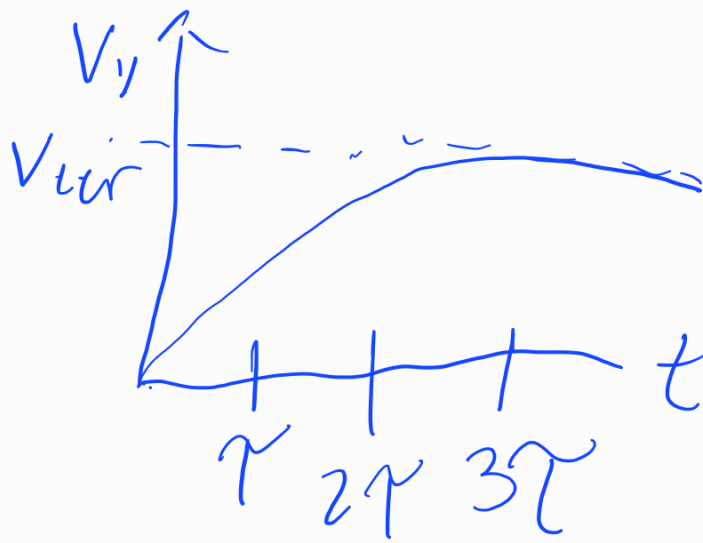
$$\text{at } t=0, A = v_{y0} - V_{ter}$$

$$\therefore v_y(t) = V_{ter} + (v_{y0} - V_{ter})e^{-t/\tau}$$

$$= v_{y0}e^{-t/\tau} + V_{ter}(1 - e^{-t/\tau})$$

as $t \rightarrow \infty$

$$V_Y(t) \rightarrow V_{ter}$$



for $V_{Y0} = 0$

$$V_Y(t) = V_{ter}(1 - e^{-t/\tau})$$

when $t = \tau$

$$V_Y = V_{ter}(1 - e^{-1}) = 0.63 V_{ter}$$

So

t	$\% V_{ter}$
0	0%
τ	63%
2τ	86%
3τ	95%

$$\rightarrow 3\tau \approx V_{ter}$$

Remember:

$$V_y(t) = \dot{y}$$

$$y(t) = \int_0^t V_y(t') dt'$$

$$= V_{ter} t + (V_{y0} - V_{ter}) \tau (1 - e^{-t/\tau})$$

$$\therefore y(t) = V_{ter} t + (V_{y0} - V_{ter}) \tau (1 - e^{-t/\tau})$$

$$x(t) = x_{\infty} (1 - e^{-t/\tau})$$

$$= V_{x0} \tau (1 - e^{-t/\tau})$$

Projectile motion in Linear Medium

Trajectory & Range in Linear Media

$$X(t) = V_{x0} \tau (1 - e^{-t/\tau})$$

$$Y(t) = (V_{y0} + V_{ter}) \tau (1 - e^{-t/\tau}) - V_{ter} \tau$$

Note: $+y$ direction is up now

by solving first for t & plugging into second

$$Y(t) = \frac{(V_{y0} + V_{ter})}{V_{x0}} X + V_{ter} \tau \ln\left(1 - \frac{X}{V_{x0} \tau}\right)$$

Trajectory Eqn

Range:

from Gen. Phys.

$$R_{vac} = \frac{2V_{x0}V_{y0}}{g} \quad [\text{no air resistance}]$$

Range R is $\propto V = 0$ So,

$$\frac{V_{y0} + V_{ter}}{V_{x0}} R + V_{ter} \ln\left(1 - \frac{R}{V_{x0} \tau}\right) = 0$$

This is a transcendental equation and has no analytical solution but can be solved numerically for a given set of Parameters

Approximate Solution:

f_{air} is usually small

$\therefore V_{ter}$ & τ are large

So $\ln(1 - \frac{R}{V_{x0}\tau})$ is small

We can expand w/ Taylor series

$$\ln(1-E) = -(E + \frac{1}{2}E^2 + \frac{1}{3}E^3 + \dots)$$

$$\left[\frac{V_{y0} + V_{ter}}{V_{x0}} \right] R - V_{ter} \tau \left[\frac{R}{V_{x0}\tau} + \frac{1}{2} \left[\frac{R}{V_{x0}\tau} \right]^2 + \frac{1}{3} \left[\frac{R}{V_{x0}\tau} \right]^3 \right] = 0$$

Every term has R

so $R=0$ is a solution

$\therefore y=0$ @ $x=0$ but this

is uninteresting

Expanding, cancelling terms
and rearranging we get:

$$R = \frac{2V_{x0}V_{y0}}{g} - \underbrace{\frac{2}{3V_{x0}} R^2}_{\text{very small}}$$
$$R < R_{vac}$$

First Approximation

$$R \approx \frac{2V_{x0}V_{y0}}{g} = R_{vac}$$

if $R \approx R_{vac}$

$$V_{ter} = \tau g$$

$$R \approx R_{vac} - \frac{2}{3V_{x0}\tau} (R_{vac})^2$$

$$= R_{vac} \left(1 - \frac{4}{3} \frac{V_{y0}}{V_{ter}} \right)$$

$$\text{If } \frac{V_{y0}}{V_{ter}} \ll 1 \quad R = R_{vac}$$

